

Lecture 2 Introduction to Linux Notes

1. What is an Operating System?

An operating system (OS) is software that manages a computer's hardware and software. It allows users to run applications and interact with the computer. Examples include Windows, macOS, and Linux.

2. What is a kernel?

A kernel is the core part of an operating system. It manages hardware resources such as the CPU, memory, and devices. It also helps software communicate with the hardware.

3. Which other parts aside from the kernel identify an OS?

Other important parts of an operating system include:

- **Shell:** Allows users to interact with the operating system.
- **File System:** Organizes and stores files and folders.
- **System Utilities:** Help manage and maintain the computer.
- **User Interface:** Allows users to interact with the computer using commands or graphics.
- **System Libraries:** Provide functions used by applications.

4. What is Linux?

Linux is a free and open-source operating system kernel created by Linus Torvalds in 1991. It is used in computers, servers, smartphones, and other devices. The word Linux is also commonly used to describe operating systems built around the Linux kernel.

5. What is a Linux distribution?

A Linux distribution is an operating system that combines the Linux kernel with other software and tools. Different distributions are designed for different purposes. Examples include Debian, Ubuntu, Fedora, and Linux Mint.

6. List at least 4 Linux characteristics:

1. **Open Source:** Users can view, modify, and share its source code under its licenses.
2. **Multiuser:** Multiple users can use the same system.
3. **Multitasking:** Linux can run multiple programs at the same time.
4. **Secure:** Linux provides user permissions and other security features.
5. **Portable:** Linux runs on many different types of hardware.
6. **Stable:** Linux is known for its reliability, especially on servers.

7. What is Debian?

Debian is a free and open-source Linux distribution maintained by a community of developers. It is known for its stability, security, and reliability. Debian uses the APT package manager to install and update software. Ubuntu is one of the Linux distributions based on Debian.

8. List and define the different types of licensing agreements

- **Proprietary License:** Software owned by a company or individual. Users have restrictions on copying, modifying, and sharing it.
- **Open-Source License:** Allows users to access, study, modify, and distribute source code under specific conditions.
- **Free Software License:** Protects users' freedoms to run, study, modify, and share software.
- **Copyleft License:** Requires redistributed modified versions to follow the same or compatible licensing rules. The GNU GPL is an example.
- **Permissive License:** Allows users to modify and redistribute software with relatively few restrictions. MIT and BSD are examples.
- **Public Domain:** Software that is not protected by copyright, allowing people to use, modify, and distribute it without copyright restrictions.
- **Freeware:** Software available at no cost but usually does not allow users to modify its source code.
- **Shareware:** Software distributed for free evaluation, often with limited features or a trial period.

9. What is Free Software? Define the 4 freedoms.

Free Software is software that gives users the freedom to run, study, modify, and share it. Free refers to freedom, not necessarily price.

The four essential freedoms are:

1. **Freedom 0:** The freedom to run the program for any purpose.
2. **Freedom 1:** The freedom to study how the program works and change it to meet your needs.
3. **Freedom 2:** The freedom to redistribute copies to help others.
4. **Freedom 3:** The freedom to distribute modified versions so others can benefit from improvements.

Access to the source code is necessary for Freedoms 1 and 3.

10. What is virtualization?

Virtualization is technology that allows multiple virtual computers to run on one physical computer. Each virtual machine can have its own operating system and software. Virtualization is useful for testing, learning, and running different operating systems without needing separate physical computers.

For example, I use UTM on my Mac to run Debian Linux in a virtual machine for my CIS 106 class.